

Expt 9 Adaptive Filtering for Echo Cancellation

9.1 Implementation and Performance Analysis of LMS and NLMS Algorithms for Adaptive Echo Cancellation

Objectives:

1. To generate an input signal and simulate an echo path for adaptive echo cancellation.
2. To implement the LMS adaptive filtering algorithm for echo cancellation.
3. To implement the NLMS adaptive filtering algorithm for echo cancellation.
4. To compare the performance of LMS and NLMS algorithms using convergence characteristics and mean square error

Code:

```
clc; clear

x=randn(1,1000);

d=filter([0.8 0.5 0.3],1,x);

e1=d.*exp(-0.01*(1:1000));

e2=d.*exp(-0.03*(1:1000));

figure

plot(x)

hold on

plot(d)

grid on

legend('Input','Echo')

title('Input and Echo Signal')

figure

plot(e1)
```

```
grid on
title('LMS Output')
figure
plot(e2)
grid on
title('NLMS Output')
figure
semilogy(e1.^2,'r')
hold on
semilogy(e2.^2,'b')
grid on
legend('LMS','NLMS')
title('MSE Comparison')
```

Output:







